DGMS 2019 • SECTION L: ENVIRONMENTAL & MARINE GEOCHEMISTRY

Question 87: Ocean Acidification & Seawater Carbonate Chemistry

Official Examination Guide • CO2 Ingress, Seawater pH Drop & Carbonate Ion Depletion in Marine Calcifiers

OFFICIAL EXAMINATION QUESTION

Carbon dioxide emissions from land entering the oceans:

1. Make oceanic water acidic

2. Make oceanic water basic

3. Help some marine organisms to make their carbonate shells by drawing carbonate from the seawater

Choose the correct option:

(a) 1 only CORRECT ANSWER	(b) 2 and 3
(c) 2 only	(d) 1 and 3

1. Executive Summary & Core Geochemical Principle

The dissolution of anthropogenic carbon dioxide (CO_2) emissions into the global oceans triggers fundamental shifts in the marine carbonate equilibrium system (William M. White, *Geochemistry*, Chapter 6):

- **Seawater Acidification (Statement 1 is TRUE):** Atmospheric CO_2 dissolves into surface seawater, forming aqueous carbonic acid ($

\text{H}_2\text{CO}_3$), which rapidly dissociates to produce hydrogen ions (H^+) and bicarbonate (HCO_3^-):

```
CO₂(g) ⇌ CO₂(aq)
CO₂(aq) + H₂O ⇌ H₂CO₃ ⇌ H⁺ + HCO₃⁻
```

The release of free hydrogen ions (H^+) directly lowers the pH of seawater (a phenomenon termed **Ocean Acidification**). Surface ocean pH has already decreased by ~0.1 units since the industrial revolution (representing an approximate 30% increase in $[\text{H}^+]$ activity). Statement 1 is therefore correct, while Statement 2 ("Make oceanic water basic") is completely false.

- **Carbonate Ion Depletion & Calcification Impairment (Statement 3 is FALSE):** A common misconception is that adding CO_2 adds carbonate for shells. The excess free H^+ ions react with ambient dissolved carbonate ions (CO_3^{2-}), converting them into bicarbonate:

```
H⁺ + CO₃²⁻ ⇌ HCO₃⁻
```

This reaction **consumes and depletes** the bioavailable carbonate ions (CO_3^{2-})!
The saturation state of calcium carbonate (Ω) is given by:

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Ω = [Ca²⁺][CO₃²⁻] / K'sp
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As $[\text{CO}_3^{2-}]$ plummets, Ω decreases toward undersaturation ($\Omega < 1$). Consequently, calcifying marine organisms (corals, pteropods, coccolithophores, foraminifera, mollusks) find it **substantially harder** to precipitate their protective shells (CaCO_3), and existing shells undergo chemical dissolution. Statement 3 is thus completely contrary to scientific fact!

Therefore, only Statement 1 is correct, making **(a) 1 only** the definitive answer.



Figure 87.1: Chemical cascade of ocean acidification illustrating the generation of H+ ions, drop in seawater pH, and consumption of carbonate ions required for marine shell calcification (Grounded in White Geochemistry and Zeebe & Wolf-Gladrow).

2. Carbonate System Species Distribution with Decreasing pH

Chemical Species	Chemical Formula	Response to Ocean Acidification (CO_2 Ingress)	Impact on Marine Ecosystem
Hydrogen Ion	H^+	**Increases significantly (~30% rise)**	Drives down ocean pH (from ~8.25 to ~8.14); increases oceanic acidity.
		Increases	Formed directly from

Chemical Species	Chemical Formula	Response to Ocean Acidification (CO_2 Ingress)	Impact on Marine Ecosystem
Carbonic Acid	H_2CO_3^*		dissolved gas: $\text{CO}_2\text{(aq)} + \text{H}_2\text{O}$.
Bicarbonate Ion	HCO_3^-	**Increases**	Becomes the heavily dominant dissolved inorganic carbon (DIC) species.
Carbonate Ion	CO_3^{2-}	**Decreases drastically (Depleted!)**	**Deprived from shell-builders!** Lowers calcite & aragonite saturation (Ω).
Calcium Carbonate Saturation	$\Omega_{\text{arag}}, \Omega_{\text{calc}}$	**Decreases toward < 1**	Causes shell dissolution, coral reef bleaching, and failure of larval calcification.

3. Evaluation of Options

- **Statement 1: CORRECT.** Ingress of CO_2 generates H^+ ions, lowering seawater pH and making the ocean more acidic.

- **Statement 2: INCORRECT.** CO_2 acts as an acid gas, lowering pH; it never makes water basic.
- **Statement 3: INCORRECT.** CO_2 consumes carbonate ions ($\text{H}^+ + \text{CO}_3^{2-} \rightarrow \text{HCO}_3^-$), depleting the carbonate necessary for shell synthesis.

> **Key Takeaway:** CO_2 dissolution releases H^+ (acidification) and consumes CO_3^{2-} (depleting shell-building minerals). Hence, only Statement 1 is correct: **(a) 1 only**.

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